

AKSHAT SHARMA

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PROFESSIONAL EXPERIENCE

Neel Data Pro IT Solutions Pvt Ltd

Data Engineer | Data Engineering Intern

July 2021 - July 2024

Greater Noida, India

- Engineered ETL pipelines in Python and SQL processing 2.5 GB of daily tourism data, consolidating 6+ source systems into Google BigQuery as the centralized analytical warehouse and cutting report generation from 4 hours to 30 minutes for 15+ government stakeholders.
- Migrated 10M+ records from legacy systems into BigQuery, standardized metadata fields, and reconciled row counts against sources to confirm zero data loss at cutover.
- Developed complex SQL in BigQuery for transformation, reconciliation, and downstream reporting, delivering curated datasets analysts queried directly for segment analysis.
- Implemented validation checks at load boundaries, row counts, null thresholds, schema conformance, and referential integrity, catching data defects before they reached stakeholder reports.
- Owned the reporting pipeline end-to-end from ingestion through delivery scheduling, failure triage, and reprocessing, and partnered with 3 cross-functional teams to resolve production bottlenecks, improving data integrity ~25% and cutting report discrepancies 30%.
- Analyzed 500K+ records in Python and pandas and built 8+ Tableau dashboards tracking segment KPIs, informing 4 strategic funding decisions and raising stakeholder engagement 40%.
- Authored pipeline runbooks and a data dictionary for the warehouse schema, establishing documented data governance that let analysts self-serve on schema questions.

PROJECTS & OUTSIDE EXPERIENCE

SoftCart E-commerce Data Platform with AI-Powered Text-to-SQL | Airflow, DuckDB, FastAPI, Docker, Ollama

- Architected an end-to-end data platform that ingests MySQL (OLTP) transactions and MongoDB (catalog) documents through a PostgreSQL staging layer into a DuckDB Kimball star schema, with 2 fact tables at the order-item grain and 7 conformed dimensions with surrogate keys.
- Implemented the ETL layer in Python, SQLAlchemy, and pandas as discrete generation, ingestion, staging, and transformation services, seeding the platform with referentially consistent synthetic data generated in Faker.
- Orchestrated the pipeline in Apache Airflow with per-task retries and two blocking data-quality gates enforcing accuracy, completeness, consistency, timeliness, and uniqueness via a pytest suite that fails the run on violation.
- Integrated a local Ollama (Qwen) natural-language-to-SQL endpoint behind a layered safety validator, single-statement SELECT enforcement, keyword denylist, table allowlist, injected LIMIT, read-only connection, validated by a pytest security suite covering injection patterns.
- Served the warehouse through a FastAPI analytics API to a Streamlit and Plotly dashboard, with Loguru structured logging and Docker Compose containerization that cut contributor setup from ~2 hours to under 10 minutes.

LangChain | Open Source Contribution

Open Source Contributor

- Diagnosed a sync/async streaming inconsistency in LangChain's Perplexity integration that corrupted response metadata across streamed chunks. Proposed the original fix in PR #38796; the diagnosis and fix were subsequently credited in merged PR #39924 and incorporated upstream.
- [Link to project](#)

YouTube Trending Data Pipeline | AWS, Python, PySpark, S3, Glue, Step Functions, Athena

- Built a cloud-native ETL pipeline on AWS ingesting YouTube Data API v3 trending data across 10 regions alongside historical Kaggle datasets, provisioning S3, Glue, Lambda, and Step Functions resources programmatically with Boto3 and landing raw files under Hive-style region partitioning.
- Designed a Bronze-Silver-Gold medallion architecture on Snappy-compressed Parquet in S3, with Glue PySpark jobs unifying JSON API and CSV schemas, deduplicating, and deriving engagement metrics into three region-partitioned Gold tables registered in the Glue Data Catalog.
- Enforced data quality at two pipeline boundaries: a pre-ingestion Lambda and a pre-Gold gate validating row-count floors, null thresholds on critical columns, schema conformance, and 48-hour freshness, halting Gold promotion and triggering SNS alerts on failure.
- Orchestrated the pipeline in Step Functions on a 6-hour EventBridge schedule with parallel Silver transformations, three-attempt exponential-backoff retries, quality-based branching, CloudWatch logging, and SNS notification at each stage.
- Delivered an Amazon QuickSight campaign dashboard over four Athena SPICE datasets answering marketing questions on regional targeting, category momentum, channel partner consistency, and publish timing, with a linked region filter and scheduled daily refresh.

EDUCATION

University of Massachusetts - Boston

Master of Science (MS), Business Analytics

- Major: AI and Data Analytics

September 2024 - May 2026

GPA: 4.0

Chitkara University

Bachelor's, Computer Science

August 2019 - August 2023

GPA: 3.74

SKILLS

- Programming & Databases:** Python, SQL, Bash/Shell, PostgreSQL, MySQL, MongoDB
- Cloud & Data Warehousing:** Google BigQuery, AWS (Glue, Lambda, S3, Athena, Step Functions, EventBridge, SNS, CloudWatch, EC2), Snowflake, Databricks, DuckDB
- Data Engineering & Orchestration:** ETL/ELT Pipeline Development, Apache Airflow, Apache Spark (PySpark), Data Quality Testing (pytest), Dimensional (Kimball) Modeling, Medallion Architecture, Data Governance & Documentation
- AI & Machine Learning:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Text-to-SQL, Ollama, OpenAI API, Vector Databases (FAISS), Prompt Engineering
- API, BI & DevOps:** FastAPI, REST APIs, SQLAlchemy, Tableau, Google Looker Studio, Streamlit, Docker, Git, Linux/Unix

CERTIFICATIONS

IBM Data Engineering Professional Certificate (Coursera) | Snowflake University Platform Skills Badge | AI Agents Fundamentals (Hugging Face)